

Automating the Processing & Analysis of Calibration Data for Solid-Contact Ion-Selective Electrodes

From raw multi-electrode traces to machine-learned concentration prediction

Yashkumar Shailesh Dodiya

Indian Institute of Technology Indore

Prof. Jennifer L. Schaefer Prof. Nosang V. Myung

Chemical & Biomolecular Engineering

Background



Water Quality Monitoring

- Accurate ion concentration measurement is essential

Ca^{2+} K^{+} Na^{+} Mg^{2+}



SC-ISE Sensors

- Low-cost
- Reliable
- Real-time monitoring



Calibration Required

- Convert sensor potential → ion concentration
- Estimate slope, intercept and linear range



Challenges

- Manual and repetitive analysis
- Time-consuming for multiple sensors
- User-dependent results



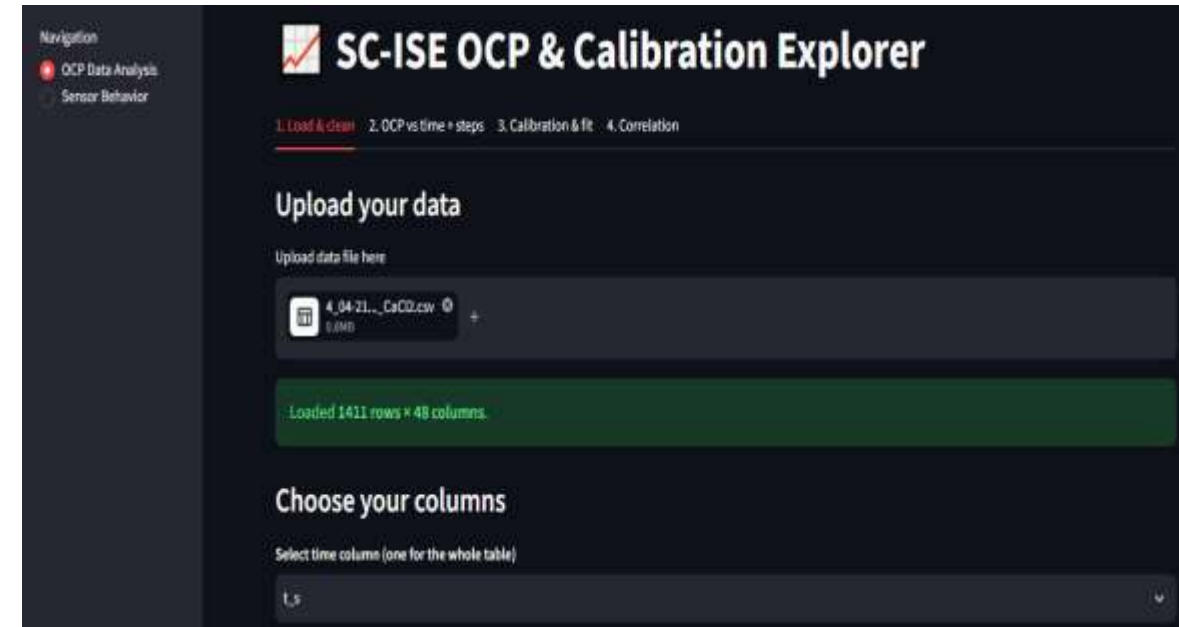
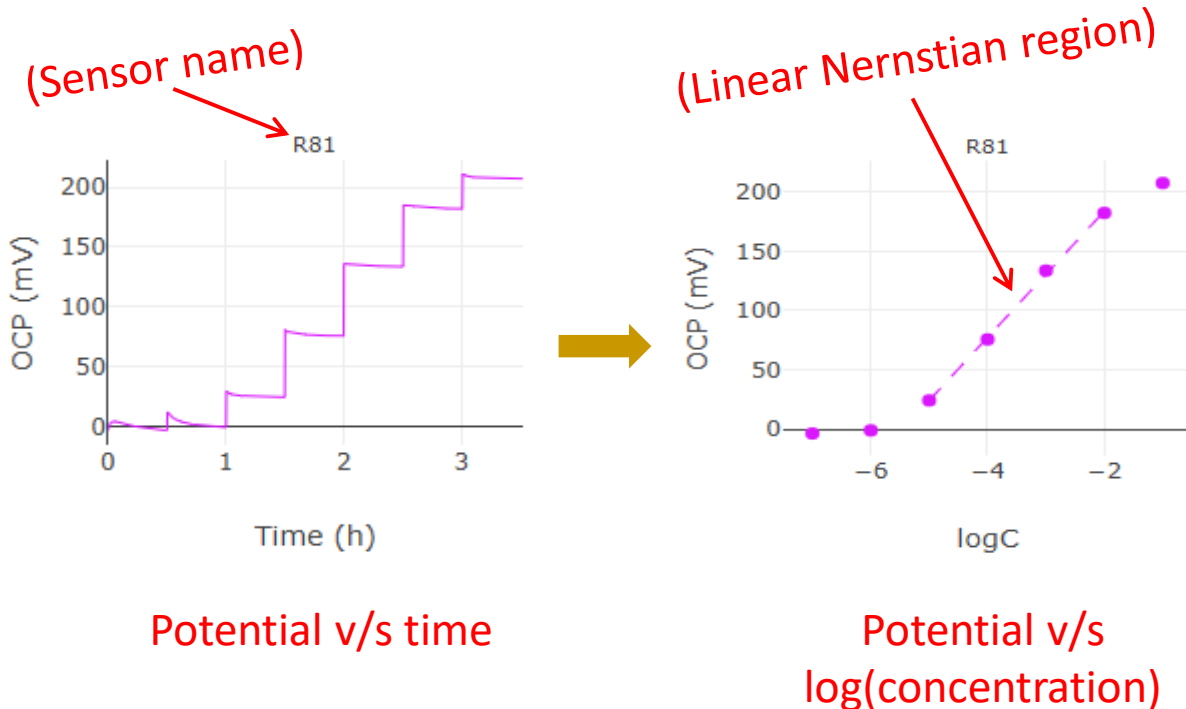
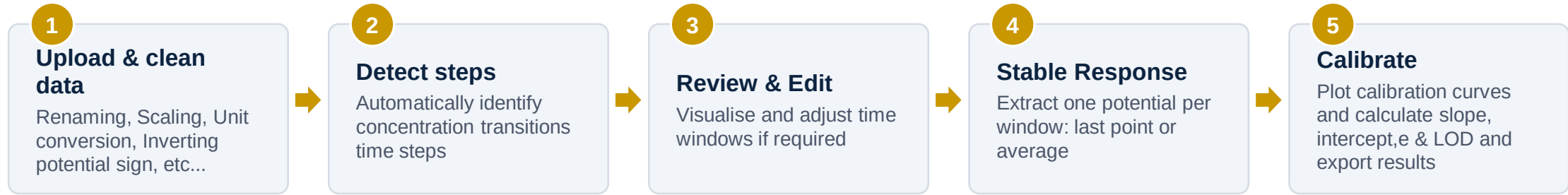
Our Solution

- Automated Python + Streamlit application
- Faster, reproducible and user-friendly analysis
- Train ML model to predict concentration



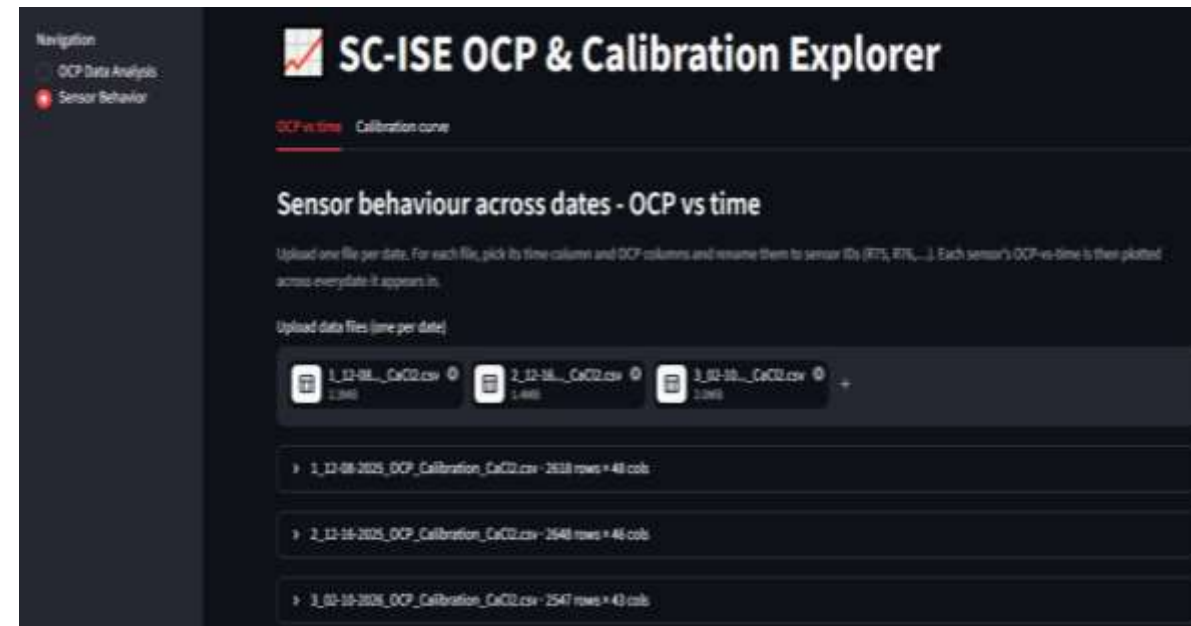
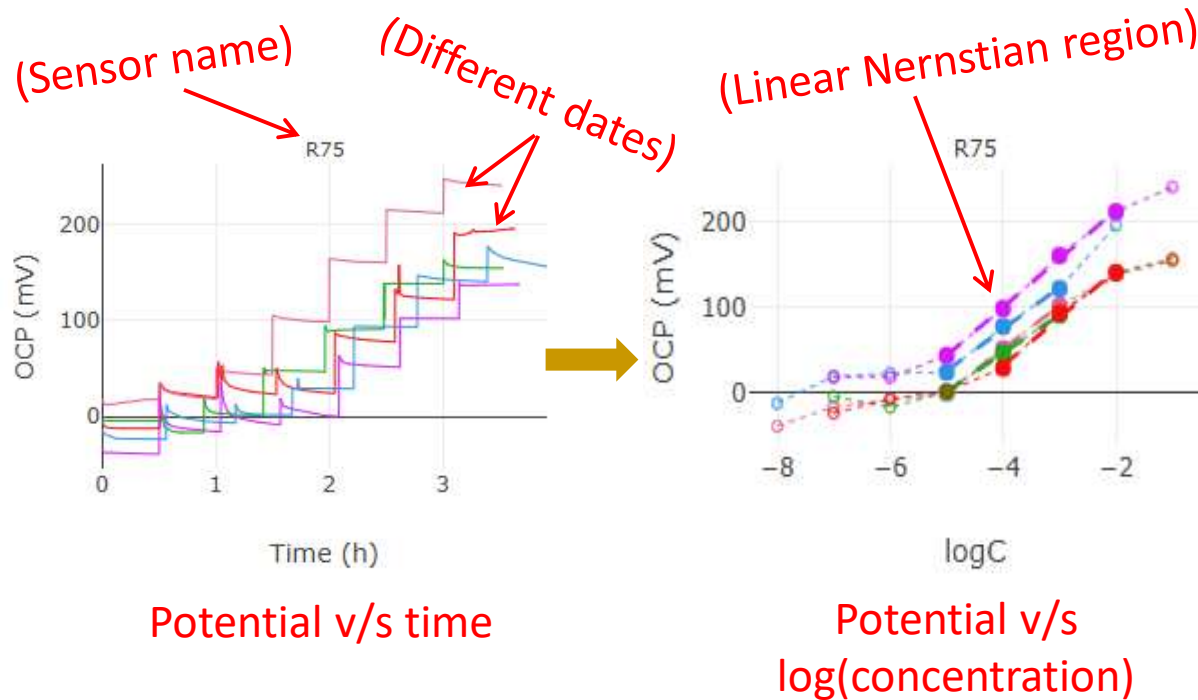
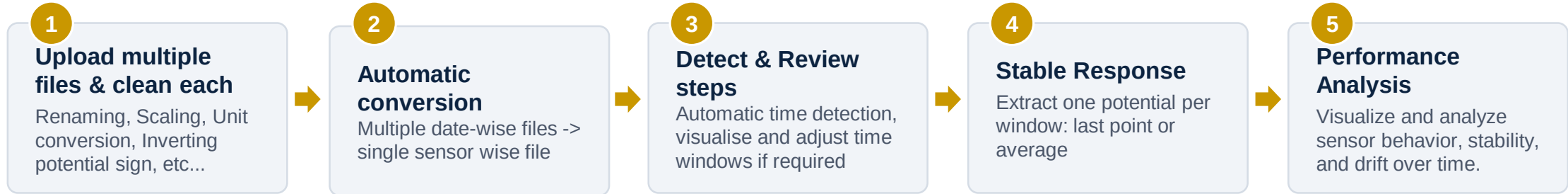
Approach : two automated pipelines in one app

Pipeline 1 : Date-wise calibration analysis



Approach : two automated pipelines in one app

Pipeline 2 : Tracking behavior of sensor over time



Results : what the automated workflow delivers

5

calibration runs, Dec 2025 – Jun 2026

44

electrodes processed

10^{-8} - 10^{-1} M

concentration range covered

-
- Automatic processing → calibration data → raw potentiometric data → slope, intercept and R^2 for every electrode, no manual curve-picking; reproducible, user-independent results
 - Sensor-wise tracking → date-wise to sensor-wise reorganization lets the pipeline follow each electrode over time
 - Minutes, not hours → no-code Streamlit app; a full multi-electrode dataset analyzed in mins instead of hours

Results : predicting concentration from response

Dataset

- One row = one concentration window of one electrode on one particular date
- Label = log base 10 of concentration
- total **573** data points

16 Features extracted per window

- mean, median, max, min, last, root mean square (absolute potential)
- range, std, max slope, min slope, mean slope, drift, mean absolute deviation, median absolute deviation (shape)
- delta previous, delta next (transition amongst concentration)

Six ML models: Ridge Regressor, k-nearest neighbours, Support Vector Regressor, random forest, extremely randomised trees, histogram gradient boosting

Training the model

- Hold out whole electrodes, and hold out whole runs => the two ways the model would actually be used: an unseen electrode, and a new fabrication batch on a new day
- A random split would leak as windows from one electrode in one run are strongly correlated

Held out by electrode

Model	R ²	Mean absolute error	Root mean square error
ExtraTrees	0.938	0.280	0.477
HistGB	0.925	0.351	0.525
RF	0.925	0.316	0.527
SVR	0.919	0.372	0.546
kNN	0.912	0.306	0.570
Ridge	0.825	0.631	0.804

Held out by run (file)

Model	R ²	Mean absolute error	Root mean square error
ExtraTrees	0.875	0.475	0.679
kNN	0.871	0.439	0.690
SVR	0.864	0.548	0.709
HistGB	0.862	0.495	0.712
RF	0.856	0.493	0.727
Ridge	0.814	0.654	0.827

Summary & Next Steps

1

Automated date-wise analysis

Step detection, reviewable switch times, windowed potentials, calibration curves, downloading & storing results

2

Sensor-wise histories

The same runs restructured per electrode, exposing drift and reproducibility

3

One app, plus a ML model

Both pipelines in a Streamlit tool.
Trained a concentration-prediction model

Next steps

- Collect more & more data and train model on it to generalise it more
- Extend beyond CaCl_2 to NaCl , KCl and MgCl_2 calibrations
- Connect the pipeline to the automated flow-injection platform
- Predict from a live trace, not only from a finished run

Acknowledgements

Prof. Jennifer L. Schaefer, Prof. Nosang V. Myung, Phd Govinda Devkota, Raul Chavez and the whole group for their mentorship and providing me with data.

Also thanks to Heidi, Derek and NDnano for providing me this opportunity.